Origins of Matrix/Vector and Matrix/Matrix Multiplication

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1 Generalizing $a\,x = b$

Why is matrix/vector and matrix/matrix multiplication defined the way it is? One motivation is to try to extend the (albeit trivial) solution of linear equations in 1-dimension to n dimensions. To do this, we go through the solution of the 1-dimensional case in careful detail.

The scalar problem is:

$$a\,x = b \tag{1}$$

Here is a very explicit solution keeping an eye towards generalization.

$$\begin{aligned}
a\,x &= b && && (2)\\
a^{-1}(a\,x) &= a^{-1}b && \text{(Multiply by Inverse)} && (3)\\
(a^{-1}a)x &= a^{-1}b && \text{(Use Associativity of multiplication)} && (4)\\
1\,x &= a^{-1}b && \text{(What Inverse multiplication does)} && (5)\\
x &= a^{-1}b && \text{(What Identity multiplication does)} && (6)
\end{aligned}$$

Keep this five-line ritual in view; it is the engine of the whole paper. The strategic move is one the reader has used since grade school: *isolate and invert* – get the unknown under a single operation, then apply that operation's inverse. Our plan is to play the *same game on a bigger board*: replace the numbers by vectors and rectangular arrays of coefficients, and then ask what "multiply," "associate," "identity," and "inverse" must *mean* for the five lines to remain legal. The payoff of this approach is that the definitions of matrix/vector and matrix/matrix multiplication will not be conventions handed down to be memorized; each will be *forced*.

The multi-dimensional problem has a lot more variables and coefficients. To start, we need to be more systematic about the naming of these coefficients. With this in mind, the multi-dimensional linear problem can be written:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ \cdots \qquad\qquad &= \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned}$$

Here, a_{ij} is the coefficient in the i^{th} row and j^{th} column.

To make this look like the 1-dimensional case, we need to think of the b's as a single unit. Our single unit will be the *vector* of the b's.

The multi-dimensional case can be rewritten in vector terms as:

$$\begin{bmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{7}$$

We can replace the tedious list of sums on the left hand side of the above with standard mathematical summing notation:

$$\begin{bmatrix} \sum_{j=1}^{n} a_{1j}x_j \\ \sum_{j=1}^{n} a_{2j}x_j \\ \vdots \\ \sum_{j=1}^{n} a_{nj}x_j \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{8}$$

Two things need to be done: treat the x's as a unit – as we did with the b's – and separate the a coefficients from the x's. This must be done formally while preserving the meaning of the original problem. This is done in the most natural of ways:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{9}$$

Taking the rectangular collection (matrix) of a's to be represented by A; the (vector) collection of x's to be $\mathbf{x}$; and the collection of the b's to be $\mathbf{b}$; we may write (9) as:[1]

$$A\,\mathbf{x} = \mathbf{b} \tag{10}$$

Here, A represents the "matrix" of the a's, with $A_{i,j} = a_{ij}$; meaning, that the i^{th} row and j^{th} column of A is a_{ij}. This now looks like the scalar problem, (1). Just as one writes $a\,x = b$, interpreting the proximity of a and x to mean multiplication, we want the proximity of A and $\mathbf{x}$ to mean matrix/vector multiplication – something we need to work out.

[1]In general, we use uppercase italic letters for matrices; lowercase bold roman letters for vectors; and lowercase italic letters for scalars.

2 Solution of $A\mathrm{x} = \mathrm{b}$ determines Matrix Multiplication

We proceed to try solving (10) using the solution procedure used above for the scalar case. In the process, we need to:

- Vectorize the input variables, x_i, and the outputs, b_i. – **Done**.
- Define an "a" matrix. – **Done**.
- Define matrix-vector multiplication.
- Define matrix-matrix multiplication.
- Define the identity matrix.
- Define the inverse matrix.

Here, in our new notation, is what the solution outline looks like for the multi-dimensional problem:

$$\begin{aligned}
A\,\mathbf{x} &= \mathbf{b} && && (11)\\
A^{-1}(A\,\mathbf{x}) &= A^{-1}\mathbf{b} && \text{(Matrix/vector multiply the vector equation by } A \text{ Inverse)} && (12)\\
(A^{-1}A)\mathbf{x} &= A^{-1}\mathbf{b} && \text{(Use Associativity of matrix/matrix multiplication)} && (13)\\
I\,\mathbf{x} &= A^{-1}\mathbf{b} && \text{(What Inverse matrix/matrix multiplication does)} && (14)\\
\mathbf{x} &= A^{-1}\mathbf{b} && \text{(What Identity matrix/vector multiplication does)} && (15)
\end{aligned}$$

How do we make sense of this matrix/vector syntax on the left hand side of (11)? It should have the proper meaning; that is, (9) should have the same meaning as (8).

Examining (9) and the left hand side of (8), gives us our definition of matrix/vector multiplication:

$$[A\,\mathbf{x}]_i \equiv \sum_{j=1}^{n} A_{ij}x_j \tag{16}$$

That is, A acts on a vector, $\mathbf{x}$, to create a new vector, $A\,\mathbf{x}$, whose i^{th} entry is the i^{th} entry of the left hand side of (8).

Another way to write this is:

$$A\mathbf{x} = \sum_{j=1}^{n} x_j \mathbf{A}^j \tag{17}$$

Here, $\mathbf{A}^j$ is the j^{th} column vector of A.

This can be seen by writing (7) as:[2]

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{n2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{nn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \tag{18}$$

From this we can see that matrix/vector multiplication of A on $\mathbf{x}$ is a "weighted" combination of the columns of A using the elements of $\mathbf{x}$ as the weights. Notice that if we pick an "$\mathbf{x}$" with all of the weight concentrated on the j^{th} entry, then matrix/vector multiplication on this specific vector yields the j^{th} column of A, $\mathbf{A}^j$. Specifically, applying A to the special vectors, $\mathbf{e}_j$ – which are 0 everywhere except at j where they are 1 – we see that $A\,\mathbf{e}_j = \mathbf{A}^j$. Consequently, matrix/vector multiplication on this set of vectors uniquely determines A – if there is another matrix that gives the same result for these vectors with respect to matrix/vector multiplication, its columns would have to match A.

We note the following for future reference.

$$[\mathbf{A}^j]_i = A_{i,j} \tag{19}$$

That is, the i^{th} entry of the column vector, $\mathbf{A}^j$, is the i^{th} row, j^{th} column of the matrix A.

Before moving on with our outline, we note one very important property of matrix/vector multiplication: It is *linear*. By that we mean the following:[3]

$$A(\alpha\mathbf{x} + \beta\mathbf{y}) = \alpha\, A\mathbf{x} + \beta\, A\mathbf{y} \tag{20}$$

One can show this from our definition of matrix/vector multiplication which we leave to the reader. What this means in practice is that for a given vector, $\mathbf{z} = \alpha\mathbf{x} + \beta\mathbf{y}$, we can compute $A\mathbf{z}$ by computing A on the "components" of $\mathbf{z}$ and then multiplying by scalars and adding the resulting vectors. This can be an easier way to compute the *action* of a matrix on a vector in some cases. The linearity result above can be extended to any finite linear combination:

$$A\left(\sum_{j=1}^{m} c_j \mathbf{x}_j\right) = \sum_{j=1}^{m} c_j A\mathbf{x}_j \tag{21}$$

Between them, the last two observations – the $\mathbf{e}_j$ pick off columns, and linearity extends a fact about the $\mathbf{e}_j$ to a fact about every vector – form a move we will replay several times below. Call it *probe with the standard basis, then extend by linearity*. We have just used the probing half to show that a matrix is completely determined by its action on the $\mathbf{e}_j$; the same probe will shortly force the formula for matrix/matrix multiplication and pin down the identity matrix, and the extend-by-linearity half will each time upgrade what happens on the $\mathbf{e}_j$ to what happens everywhere.

[2] Here, we assume the reader has a passing knowledge of how vectors are added and multiplied by scalars.

[3] You might ask: "How did someone think up such a strange property?" But it is not strange, it is what one might hope for, as all we are saying is that matrix/vector multiplication works the same way that ordinary *scalar* multiplication does – it *distributes* through addition and scalar multiplication.

Continuing with our solution outline, matrix/matrix multiplication is needed to go from equation (12) to (13). How must this be defined? Well, we need to make sense of:

$$(A^{-1}A)\mathbf{x} = A^{-1}(A\,\mathbf{x}) \tag{22}$$

This involves the inverse of the matrix A, which we also need to define. We want associativity to hold *for all* matrix products, not just $A^{-1}A$. So we forgo what the inverse of a matrix might mean and focus on imposing the condition that matrix/matrix multiplication be associative for *all* input vectors. Specifically, the condition we impose is:

$$(B\,A)\mathbf{x} = B(A\,\mathbf{x}) \quad \forall \mathbf{x} \in \mathbf{R}^n \tag{23}$$

Notice that we are imposing what matrix/matrix multiplication is by saying how the new matrix formed, $B\,A$, acts on an *arbitrary* vector, $\mathbf{x}$. And we specify that action by the right hand side of (23), which involves only matrix/vector multiplication – something we have defined.

We want to emphasize that this is a strong condition to impose; meaning, that the requirement that this is true for *all* $\mathbf{x}$ gives us a lot to work with. For instance, if one had two $n \times n$ matrices, C and D, what could you conclude if someone told you that $C\mathbf{x} = D\mathbf{x}$ for *some* n vector, $\mathbf{x}$? Well, the answer is: not much. However, if I told you that $C\mathbf{x} = D\mathbf{x}$ *for all* $\mathbf{x}$, what could you say? I claim that tells us that $C \equiv D$; meaning that every entry in C is the same as the corresponding entry in D. We can see this by applying C and D to each of the $\mathbf{e}_j$, ($j \in [1, n]$) vectors. For each j, $C\mathbf{e}_j$ and $D\mathbf{e}_j$ pick off the j^{th} column of C and D respectively. So we see that each column of C and D must match; consequently, $C \equiv D$.

We apply this idea to (23) to find a formula for matrix/matrix multiplication. Since (23) must be true *for all* $\mathbf{x}$; in particular, it must be true for the vectors $\{\mathbf{e}_j\}_{j=1}^n$ – the probing half of our move. This gives us the following equations:

$$(B\,A)\mathbf{e}_j = B(A\,\mathbf{e}_j) \qquad j \in [1, n] \tag{24}$$

Or,

$$(B\,A)^j = B\,\mathbf{A}^j \qquad j \in [1, n] \tag{25}$$

For any given, i, the i^{th} entry of the left and right hand side is (after using the definition of matrix/vector multiplication, (16), on the right hand side)

$$\left[(B\,A)^j\right]_i = \sum_{k=1}^{n} B_{i,k} \left[\mathbf{A}^j\right]_k \quad \forall i, j \in [1, n] \tag{26}$$

Using (19) on the left and right hand sides of the previous equation we have

$$(B\,A)_{i,j} = \sum_{k=1}^{n} B_{i,k} A_{k,j} \quad \forall i, j \in [1, n] \tag{27}$$

And we are done: we have shown what the new matrix $B\,A$ is by giving every entry $(B\,A)_{i,j}$. So using the set of vectors, $\{\mathbf{e}_j\}_{j=1}^n$, we have completely determined what the matrix multiplication of $B\,A$ is in order that matrix/matrix multiplication be associative when using these vectors. Does this mean that associativity works for *arbitrary* vectors? The answer is yes, by the linearity of matrix/vector multiplication – the extend-by-linearity half of our move. Any given $\mathbf{x}$ can be written as a *linear combination*[4] of the vectors $\{\mathbf{e}_j\}_{j=1}^n$: $\mathbf{x} = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \cdots + a_n\mathbf{e}_n$. Matrix multiply the left and right hand sides by the matrix, $B\,A$. This gives

$$
\begin{aligned}
(B\,A)\mathbf{x} &= (B\,A)\left(\sum_{j=1}^{n} a_j\mathbf{e}_j\right) && \text{Replace } \mathbf{x} \text{ with its components, } \mathbf{e}_j. \\
&= \sum_{j=1}^{n} a_j(B\,A)\mathbf{e}_j && \textit{Linearity} \text{ of matrix/vector multiplication by } (B\,A) \text{ (itself a matrix).} \\
&= \sum_{j=1}^{n} a_j B(A\mathbf{e}_j) && \textit{Associativity} \text{ of matrix/matrix multiplication on components, } \mathbf{e}_j. \\
&= B\left(\sum_{j=1}^{n} a_j A\mathbf{e}_j\right) && \textit{Linearity} \text{ of matrix/vector multiplication by } B. \\
&= B\left(A\left(\sum_{j=1}^{n} a_j\mathbf{e}_j\right)\right) && \textit{Linearity} \text{ of matrix/vector multiplication by } A. \\
&= B(A\mathbf{x}) \qquad \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{28}
$$

In the above calculations, we applied our new matrix, $(B\,A)$ to an arbitrary $\mathbf{x}$, and used the *linearity* of matrix/vector multiplication and the *associativity* of matrix/matrix multiplication on the components, $\{\mathbf{e}_j\}_{j=1}^n$, to show that matrix/matrix multiplication, as we have defined it, is associative – *irrespective* of the input vector.

The next order of business is to identify an $n \times n$ matrix which serves as an identity (in terms of matrix/vector multiplication). Let us suppose that we have such a matrix and let's call it I. Then we must have $I\,\mathbf{x} = \mathbf{x} \quad \forall \mathbf{x} \in \mathbf{R}^n$. Then, in particular (probing with the standard basis once more), $I\,\mathbf{e}_j = \mathbf{e}_j \quad \forall j \in [1, n]$. However, we know that $I\,\mathbf{e}_j = \mathbf{I}^j$. Therefore, I must have the property that $\mathbf{I}^j = \mathbf{e}_j$. Consequently, I must have the form:

$$
I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \tag{29}
$$

We have shown that the only candidate for the identity matrix with respect to matrix/vector multiplication is the matrix I. That is, if there is an identity matrix, it must be the matrix of equation (29). Does it satisfy the property of being an identity matrix (again, in the matrix/vector multiplication world)? That is, do we have $I\,\mathbf{x} = \mathbf{x}$ for all $\mathbf{x} \in \mathbf{R}^n$? As before, we can express $\mathbf{x}$ as: $\mathbf{x} = \sum_{j=1}^n c_j\mathbf{e}_j$. By the *linearity* of

[4] Again, we assume the reader is familiar with expanding a vector in terms of the standard basis vectors.

matrix/vector multiplication, we have

$$\begin{aligned}
I\mathbf{x} &= I\left(\sum_{j=1}^{n} c_j \mathbf{e}_j\right) && \text{Replace } \mathbf{x} \text{ with components, } \mathbf{e}_j. \\
&= \sum_{j=1}^{n} c_j I \mathbf{e}_j && \textit{Linearity} \text{ of matrix/vector multiplication.} \\
&= \sum_{j=1}^{n} c_j \mathbf{e}_j && \textit{Identity} \text{ of } I \text{ on components, } \mathbf{e}_j. \\
&= \mathbf{x} && \text{Replace components with } \mathbf{x}.
\end{aligned} \tag{30}$$

One can show that this identity matrix, I, is also the identity operator for matrix/matrix multiplication: applying $(IA)\mathbf{e}_j = I(A\mathbf{e}_j) = A\mathbf{e}_j$ to each j shows $(IA)^j = A^j$ for all j, hence $IA = A$; the same argument on the columns of A gives $AI = A$.

The only remaining piece of the ritual is the inverse: when does a matrix A^{-1} exist, and how might one compute it? We do not attempt this here – it is genuine unfinished business, and we flag exactly where the difficulty lives: an inverse must undo A on *every* vector, and whether that is possible depends on whether the columns of A are rich enough to reach all of $\mathbf{R}^n$ (a first taste of this appears in the next section). The companion paper "What is a Determinant" builds the tool that detects the failure: a single number, the signed factor by which A scales volume, whose multiplicative property shows that if A^{-1} exists then $|A|\,|A^{-1}| = |I| = 1$ – so a matrix with determinant 0, one that flattens the unit cube, can have no inverse.

Here is the state of the dictionary between the scalar world and its vectorized cousin:

Scalar world	**Matrix world**	**Status**
$a\,x = b$	$A\,\mathbf{x} = \mathbf{b}$	the problem, repackaged
product $a\,x$	$[A\,\mathbf{x}]_i = \sum_j A_{ij} x_j$	forced: must preserve the equations
$(b\,a)\,x = b\,(a\,x)$	$(BA)\mathbf{x} = B(A\mathbf{x})$, all $\mathbf{x}$	imposed; forces $(BA)_{ij} = \sum_k B_{ik} A_{kj}$
the number 1	the matrix I	forced: $I\mathbf{e}_j = \mathbf{e}_j$ picks its columns
a^{-1} (exists iff $a \neq 0$)	A^{-1}	deferred: see "What is a Determinant"
range of a: $\{0\}$ or $\mathbf{R}$	range of A: span of its columns	next section

With the dictionary in hand, we continue with a qualitative comparison of the solution to the scalar problem, $a\,x = b$, and its vectorized cousin.

3 Qualitative Features of Solutions

We can view the multiplication of two numbers, a and x, as just that. Or, we can think of a being fixed and letting x "run through" all numbers. Here we see two cases: if $a \neq 0$, then letting x run through all of the numbers in $\mathbf{R}$ will produce all of the numbers in $\mathbf{R}$. We could think of a as an "operator" and call the set of all possible outputs, the range of a and denote it: $\mathcal{R}(a)$. There is another case: a could be zero. In this case its range is the set $\{0\}$. In the first case, with non-zero a it is clear that we can find an x to "hit" a given value b. That is, we can solve $a\,x = b$.

One can define the same concept for a matrix, A. In this case, we produce a "space" of all possible outputs of A as it acts on every input vector $\mathbf{x} \in \mathbf{R}^n$. This amounts to, using (17), the set of all *linear combinations* of the columns of A. Using this language of ranges, here is what we can say about the scalar problem: $a\,x = b$.

Unique Solution: If a^{-1} exists ($a \neq 0$), then there is a **unique** solution.

No Solution: If a^{-1} does not exist (i.e., $a = 0$) *AND* b is **not** in the range of a (that is: $b \neq 0$), then there is **no** solution.

Infinite Solutions: If a^{-1} does not exist (i.e., $a = 0$) *BUT* b is in the range of a (that is: $b = 0$), then there are an **infinite** number of solutions.

Here is the analog of this solution categorization for the multi-dimensional case: $A\,\mathbf{x} = \mathbf{b}$.

Unique Solution: If A^{-1} exists, then there is a **unique** solution.

No Solution: If A^{-1} does not exist *AND* $\mathbf{b}$ is not in the range of A (that is $\mathbf{b} \notin \mathcal{R}(A)$), then there is **no** solution.

Infinite Solutions: If A^{-1} does not exist *BUT* $\mathbf{b}$ is in the range of A (that is: $\mathbf{b} \in \mathcal{R}(A)$), then there are an **infinite** number of solutions.

4 Curtain Call

We can now answer the opening question in the paper's own words. Matrix/vector multiplication is defined the way it is because it is the only definition under which a rectangular array of coefficients can act on the unknowns-as-a-unit while *preserving the meaning* of the original system of equations. Matrix/matrix multiplication is defined the way it is because it is the only definition under which the five-line scalar ritual – multiply by the inverse, reassociate, cancel – remains legal on the bigger board: we *imposed* associativity for all inputs, probed with the standard basis, and the row-by-column recipe $(BA)_{ij} = \sum_k B_{ik}A_{kj}$ fell out. The recipe usually handed down as a convention is not one; it is what associativity looks like in coordinates. And the game itself – isolate and invert – is unchanged from the one-dimensional case; only the board grew.